

Gradient-driven discovery of Chern sectors in a high-dimensional band-structure family

Working report

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Abstract

Random sampling of a Hamiltonian parameter space can only reach a phase whose share of the volume exceeds roughly $1/N$; a walk that moves continuously through the space is instead limited by *connectivity*, and can enter a small phase by crossing into it from a neighbour. We test this on a generalized two-dimensional Chern insulator with 50 free couplings. The family is chosen because its inverse problem is open: the Chern number of a given θ is cheap to *verify*, but no construction is known that produces a θ realising a prescribed $\mathcal{C}$. The only closed form we have, $a_n = b_n = c_n = 1$, yields the perfect squares $\pm n^2$ and accounts for **18 of the 155** sectors a 10^6 -sample reference realises; combining harmonics is not additive, so the obvious composition route is closed. At equal budget a label-free gradient walk finds **+68 more sectors** than random sampling at 3×10^5 states (207 vs 139, median of five fresh seeds), a margin that **grows monotonically from +21** at 3×10^4 . Both methods saturate every volume decade a 10^6 reference resolves, so the entire measurable difference lies in the two rarest bins, where the walk leads by $1.9\times$ and $17.6\times$. Pooled over five runs it realises **283 sectors against random sampling's 160**, adds 123 that 2.5×10^6 random samples never produced, and of its 283 sectors **259 have no known closed-form representative**. All labels are computed on a converged Brillouin-zone grid ($M = 768$); we show that the $M = 96$ grid used in an earlier draft of this report is wrong on $\approx 65\%$ of points and *understated* every margin reported here.

Summary of claims

- **The target set cannot be enumerated.** $|\mathcal{C}| \leq 2H^2 = 512$ is proved; the best explicit construction reaches $H^2 = 256$ and covers only $\pm n^2$; whether the bound is attained is open. **92% of the sectors the walk realises have no known closed-form representative.** This is what makes search the only available access to them.
- **The advantage is not bought by sacrificing common phases.** Both methods saturate all four volume decades the reference resolves (33/33, 26/26, 29/29, 30/29.4). The walk pays no coverage penalty for chasing rare sectors — a real failure mode for a correlated trajectory, and one it avoids. Note that this agreement is *forced*: those bins hold 118 sectors in total and 3×10^5 random draws are expected to miss 0.2 of them, so the tie cannot by itself distinguish a different access mechanism from a better sampler.
- **More samples of the same measure do not close the gap.** Random sampling with 2.5×10^6 draws — $1.7\times$ the walk's 1.5×10^6 queries — still misses **123 sectors** the walk found, and contributes none the walk did not. This is the falsifiable form of the claim, and it uses only the standard baseline.
- **The margin grows with budget.** +21 at 3×10^4 to +68 at 3×10^5 , increasing end-to-end in all five seeds. Random sampling saturates; the walk does not.
- **Scope.** Established at 50 parameters on this model. At 20 and 32 parameters the walk loses; at 98 the seed spread makes no claim supportable. All sectors are integer Chern numbers — nothing here is a claim about phases of a new *type*, only about which of them are findable.

1 Motivation

Mapping a phase diagram means deciding, for each point of a parameter space, which phase the ground state belongs to. When the space has a handful of knobs this is done by scanning a grid. When it has tens of knobs — realistic for a material family with several competing hoppings, or a synthetic-matter platform with many independently tunable terms — a grid is impossible and the default fallback is random sampling.

Random sampling can only land in a phase whose share of parameter-space volume exceeds roughly $1/N$ for N samples. Phases occupying small volumes are reached only with vanishing probability, and in the models of interest the volume spectrum extends over many decades. An exploration method that moves *continuously* through the space is not bound by that limit: it can enter a small phase by crossing into it from a neighbour. Whether that theoretical advantage survives contact with a real objective — one that never sees a phase label — is the question this report addresses.

Machine-learned phase classification is by now standard when labels or ordered training data are available [1, 2, 3, 4]. The problem here is upstream of classification: not *how to label* the points one has, but *which points to compute* when each ground state costs something and the interesting phases occupy a vanishing fraction of the space.

2 Model

2.1 Background

A two-band Bloch Hamiltonian in two dimensions,

$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad \mathbf{d} : \text{BZ} = T^2 \rightarrow \mathbb{R}^3, \quad (1)$$

has a gapped ground state wherever $|\mathbf{d}(\mathbf{k})| \neq 0$, and the occupied band is characterised by the Chern number, minus the degree of the map $\hat{\mathbf{d}} = \mathbf{d}/|\mathbf{d}| : T^2 \rightarrow S^2$,

$$\mathcal{C} = -\frac{1}{4\pi} \int_{\text{BZ}} \hat{\mathbf{d}} \cdot \left(\partial_{k_x} \hat{\mathbf{d}} \times \partial_{k_y} \hat{\mathbf{d}} \right) d^2k \in \mathbb{Z}. \quad (2)$$

(The sign is the occupied-band convention; $+\deg \hat{\mathbf{d}}$ would label the empty band.) $\mathcal{C}$ is the TKNN invariant [5]: quantised, unable to change without closing the gap, and fixing the quantised Hall response. The canonical lattice realisations are Haldane’s model [6] and the Qi–Wu–Zhang (QWZ) model [7],

$$d_x = \sin k_x, \quad d_y = \sin k_y, \quad d_z = m + \cos k_x + \cos k_y, \quad (3)$$

which realises $\mathcal{C} = 0, \mp 1, 0$ as m passes through $-2, 0, 2$. QWZ has one parameter and three phases: far too small to be a search problem.

2.2 The generalized family used here

We keep the QWZ structure and add harmonics, which is physically the statement that hoppings extend beyond nearest neighbour:

$$\begin{aligned} d_x(\mathbf{k}; \theta) &= \sum_{n=1}^H a_n \sin(nk_x), \\ d_y(\mathbf{k}; \theta) &= \sum_{n=1}^H b_n \sin(nk_y), \\ d_z(\mathbf{k}; \theta) &= m + \sum_{n=1}^H c_n (\cos(nk_x) + \cos(nk_y)) + e \cos k_x \cos k_y, \end{aligned} \tag{4}$$

with $\theta = (a_1..a_H, b_1..b_H, c_1..c_H, m, e) \in \mathbb{R}^{3H+2}$ normalised to the unit sphere ($\mathcal{C}$ is scale invariant). $H = 16$ gives the **50-parameter** family used throughout.

2.3 Why the reachable set cannot be enumerated

This subsection states the property that makes the family a search problem rather than a lookup.

An upper bound, proved. Zeros of d_x : writing $z = e^{ik_x}$, $\sum_n a_n \sin(nk_x)$ has a numerator polynomial of degree $2H$ in z , hence at most $2H$ roots; likewise for d_y . The degree of $\hat{\mathbf{d}}$ is the signed count of preimages of any regular value, so we may take the two poles of S^2 : the $\leq 4H^2$ points with $d_x = d_y = 0$ split according to $\text{sign}(d_z)$ into N_+ mapping to the north pole and N_- to the south, and

$$|\mathcal{C}| \leq \min(N_+, N_-) \leq \frac{1}{2}(N_+ + N_-) \leq 2H^2 = 512. \tag{5}$$

The halving step is essential: the crossing count alone gives only $4H^2$.

A construction, covering 12%. Setting $a_n = b_n = c_n = 1$ for a single harmonic n and sweeping m tiles the Brillouin zone $n \times n$ and gives $\mathcal{C} = \pm n^2$ (verified for $n = 1 \dots 16$ at $m = \pm 1$, with $\mathcal{C} = 0$ at $m = \pm 3$). This is the only closed form we have. It reaches $H^2 = 256$ — a *factor of two below the bound* — and produces only perfect squares.

Composition fails. Turning on two harmonics does not add their degrees. Measured at $m = +1$: $(n_1, n_2) = (2, 3)$ gives $\mathcal{C} = 8$, not 13; $(3, 5)$ gives 9, not 34; $(1, 16)$ gives 44, not 257; $(1, 2)$ is gapless and carries no label at all. The dependence of $\mathcal{C}$ on θ is not decomposable, so the natural route from the single-harmonic family to an arbitrary target is closed.

Consequence. Of the 155 sectors realised in a 10^6 -sample Gaussian reference, **18 are of the form $\pm n^2$ and 137 are not**. Of the 283 sectors the walk realises, **24 are constructible and 259 are not**. There is no list to check off and no formula to invert: for a target such as $\mathcal{C} = 137$ we know of no way to write down a θ that realises it. Search is the only access we have, which is exactly the condition under which an exploration method is worth measuring.

2.4 Labels: what is exact and what is not

$H(\mathbf{k})$ is 2×2 , so the occupied eigenvector is closed form: there is no diagonalisation, no convergence criterion and no finite-size effect. The *label*, however, is a discretised functional. We evaluate $\mathcal{C}$ by the Fukui–Hatsugai–Suzuki lattice method [8], which is gauge invariant and integer by construction, but whose convergence condition is that the Berry flux per plaquette be resolved — not, as an earlier draft of this report asserted, that the grid resolve every harmonic of $\mathbf{d}(\mathbf{k})$.

At $H = 16$ there are up to $4H^2 = 1024$ monopole crossings to resolve. Measured against $M = 1536$ on 3000 points:

label grid	agrees with $M=1536$	mean $ \Delta\mathcal{C} $	max
$M = 96$ (earlier draft)	$\approx 35\%$	3.2	16
$M = 192$	70%	1.1	8
$M = 384$	90–93%	0.3	4
$M = 768$ (used here)	98.2%	0.06	4

Table 1: Convergence of the FHS integer at $H = 16$. The $M = 96$ grid is wrong on about two thirds of points, and its error *compresses* $|\mathcal{C}|$ toward zero.

Two facts make this recoverable rather than fatal. First, the error is **symmetric between the two methods**: at $M = 768$ the agreement with $M = 1536$ is 98.3% on the walk’s θ and 98.2% on the baseline’s; at $M = 96$ it is 36.8% and 34.5%. The grid never favoured either side. Second, correcting it *increased* every margin (§5). All numbers in this report are at $M = 768$; the residual $\approx 2\%$ disagreement with $M = 1536$ is an acknowledged uncertainty.

Validated against QWZ: $\mathcal{C} = 0, -1, +1, 0$ at $m = -3, -1, 1, 3$, reproduced by three independent implementations. That check validates the implementation at $H = 1$ and says nothing about grid convergence at $H = 16$.

2.5 No shortcut through the coordinate axes

Sampling the 50 coordinate axes (both signs) returns a single sector, $\mathcal{C} = 0$. The mechanism is not that one dominant parameter gives a trivial configuration: **98 of the 100 signed axes are gapless**, so $\mathcal{C}$ is undefined and the point is discarded; only $\pm m$ is gapped, and it gives 0. The residual advantage of a heavy-tailed measure over a Gaussian one is $1.19\times$.

3 The complementary 1D benchmark

The Chern family has an open inverse problem but a discretised label. A one-dimensional benchmark with the opposite profile — an *exact* label but a trivial inverse — is reported separately [10] and is summarised here because the two together close each other’s gaps.

After a Jordan–Wigner transformation the generalized cluster chain is the long-range Kitaev chain in class BDI, $H = i \sum_j \sum_{\alpha=0}^{d-1} c_{\alpha} a_j b_{j+\alpha}$, $\mathbb{Z}$ -classified by the winding number of $f(k) = \sum_{\alpha} c_{\alpha} e^{ik\alpha}$. Labels there are computed by **exact companion-matrix root counting**, so they carry *no* lattice-resolution error of the kind quantified in §2.4. Against a rotation-invariant Gaussian baseline, over a 10×4 grid of $d \in [20, 200]$ and budgets 10^4 – 3×10^5 , the walk finds 5–25 more phases at equal budget; in 36 of 40 cells its worst seed beats the baseline’s best realisation; and the walk at 10^4 solves exceeds uniform sampling at 3×10^5 — a measured $30\times$ query advantage.

Two cautions apply, and they are why that model cannot carry the discovery claim alone. First, **its inverse problem is trivial**: e_α realises $\omega = \alpha$, so all d sectors have one-line representatives and nothing is discovered in the sense of §2.3. Second, that same structure makes the benchmark sensitive to the sampling measure — coordinate-axis or heavy-tailed sampling obtains phases for free, and an advantage measured against those baselines does not survive a hidden orthogonal rotation of the couplings. Both are handled in [10] by using the rotation-invariant measure throughout and by decomposing the margin by starting point: the basis-free component (random start) is +3.5, +3.5, +6.5, +11.0 at $d = 60, 100, 140, 200$ — positive, growing with d , and at $d = 200$ larger than the bonus from starting at the trivial limit.

The division of labour. The 1D model shows the efficiency advantage is neither a lattice artefact nor a basis artefact. The Chern model shows it delivers sectors that cannot be written down. Neither claim is available from the other model.

4 Method

4.1 What the explorer does

The explorer is a gradient walk on the parameter sphere. At θ it evaluates the ground state, encodes it, and steps to increase a novelty objective:

$$\mathcal{L}(\theta) = -\|z(\theta) - \mu\|^2 + w \sum_{i \in \text{trail}} \exp\left(-\frac{\|\theta - \theta_i\|^2}{s^2}\right), \quad (6)$$

where z is the latent code of a small autoencoder trained on a cloud of states around the run’s starting point, μ is that cloud’s latent centroid, and the second term repels the walk from its own recent trajectory. Updates use Adam with cosine-decayed step size, gradient-norm clipping, and radial projection back onto the sphere.

The state representation handed to the autoencoder is the normalised field $\hat{\mathbf{d}}(\mathbf{k})$ on a Brillouin-zone grid, flattened — the occupied band projector up to constants, $P_-(\mathbf{k}) = \frac{1}{2}(1 - \hat{\mathbf{d}} \cdot \boldsymbol{\sigma})$.

4.2 Implementation

	setting	value at $H = 16$
input to the AE	$\hat{\mathbf{d}}(\mathbf{k})$ on an $M \times M$ grid, flattened, rescaled to unit norm	$M = 32$, 3072 numbers
autoencoder	[3072, 32, 16, 32, 3072], latent on the unit sphere	$\approx 0.2\text{M}$ params
AE training	fidelity loss $1 - \langle \hat{x}, x \rangle ^2$, Adam 10^{-3}	1500 epochs
AE training set	states in a Gaussian cloud of radius r about the current point	30 states
label grid	separate, converged BZ grid for Eq. (2)	768 $\times$ 768

A known defect of the encoder grid. At $H = 16$ the AE grid satisfies $M = 32 = 2H$ exactly, so $\sin(Hk)$ evaluated at $k = 2\pi j/M$ vanishes identically. **a_{16} and b_{16} are therefore invisible to the autoencoder**: perturbing a_{16} by 0.3 changes the AE input by 2×10^{-15} while moving $\mathcal{C}$ from -11 to $+1$. Those two of the fifty directions receive *exactly zero* gradient from the first term of Eq. (6) and

move only under trail repulsion and the sphere projection. The results below are therefore obtained with 48 effective search directions; we did not tune M , and whether repairing it helps is untested.

Nothing in Eq. (6) sees a phase label. The objective is built from the state and from the trajectory’s own history. The walk starts at $\theta = (0, \dots, 0, m=1, 0)$, which is $\mathcal{C} = 0$ — a choice requiring no knowledge of the phase diagram (“turn on only the mass term”) and which is, if anything, disadvantageous: the Chern ladder runs in both directions from 0, so the walk must climb while random sampling lands directly at large $|\mathcal{C}|$.

4.3 Protocol

- **Baseline:** plain Gaussian sampling of θ , no measure engineering, recomputed in-job at exactly the explorer’s budget with the identical label function and accept criterion.
- **Budget accounting.** One unit per θ at which the state is evaluated, for both sides. Bootstrap samples are charged. The **lazy** setting (one gradient solve per 5 parameter steps) grants no free observations: intermediate steps reuse the stale gradient and are never evaluated or labelled. Instrumented re-runs of all five production seeds recorded **exactly 300,000 paid θ each, overshoot zero**.
- **What the walk gets for free.** Per unit of budget it also obtains ∇_{θ} of the state by automatic differentiation, and evaluates the state on two grids ($M = 32$ for the gradient, $M = 768$ for the label) where the baseline evaluates one. Both are free for a 2×2 Bloch model. In a setting where the budget proxies real solver calls, a parameter-space gradient is *not* one solve, and the equal-budget comparison should be read with that caveat. This is the load-bearing modelling assumption of the comparison.
- **Hyperparameters** were selected by a Hyperband bracket [9] ($27 \rightarrow 9 \rightarrow 3 \rightarrow 1$, $\eta = 3$) over eight parameters sampled log-uniformly; because an exploration run is naturally *continuable*, a promoted trial resumes its walk instead of restarting.
- **Selection bias.** The bracket reports the maximum of 27 trials each measured once; run-to-run standard deviation of the walk’s count is 9–26 depending on budget (25.7 at 3×10^5), comparable to the effect, so the bracket estimate is biased upward. One bracket winner fell from +9 in-bracket to **−1** on fresh seeds. All results below use seed family 200–204, disjoint from the bracket’s. **An earlier draft also reported a 67,500-budget row; that row was the set on which the winning configuration was chosen among three finalists and is omitted here as selection-contaminated.**

5 Results

5.1 Coverage at equal budget

The four budgets are nested prefixes of the same runs, so the comparison below is paired by construction.

5.2 Where the advantage lives

Resolving the same comparison by the volume each sector occupies under the Gaussian measure, using a 10^6 -sample reference relabelled at $M = 768$ (155 sectors):

budget	explorer	random	difference	per-seed differences
3×10^4	130	109	+21	11, 13, 20, 28, 61
6×10^4	150	118	+32	16, 21, 32, 35, 76
1.2×10^5	175	124	+51	29, 38, 50, 51, 101
1.8×10^5	189	129	+60	27, 41, 62, 66, 111
2.4×10^5	199	135	+64	32, 45, 68, 68, 120
3×10^5	207	139	+68	34, 48, 68, 74, 125

Table 2: 50 parameters, converged labels ($M = 768$), median of *five* fresh seeds; six of the ten measured 3×10^4 -spaced budgets shown. Every seed is positive at every budget and every seed’s margin is larger at 3×10^5 than at 3×10^4 . At 3×10^4 granularity the margin is strictly monotone in 3 of 5 seeds; the other two fluctuate by at most 3.

seed	explorer	random	difference	explorer’s $\mathcal{C}$ range
200	181	133	+48	[−109, 120]
201	262	137	+125	[−192, 188]
202	213	139	+74	[−180, 165]
203	207	139	+68	[−152, 172]
204	175	141	+34	[−105, 157]

Table 3: Per-seed detail at 3×10^5 . Seed-to-seed spread is large ($\sigma = 25.7$ for the explorer), which is why five seeds and min–max bands, not standard errors, are reported throughout.

The four upper bins are *saturated*: they contain 118 sectors in total and a 3×10^5 random sample is expected to miss 0.2 of them. Both methods therefore find essentially all of them, and this agreement cannot by itself discriminate between a different access mechanism and a better sampler — no method can win where none can lose. What it does establish is a real negative: a correlated trajectory could easily have traded common-phase coverage for rare finds, and this one does not.

The discriminating statement is the pooled comparison at unequal budget in the walk’s disfavour:

Random sampling was given $1.7\times$ the walk’s budget and still **contributes no sector the walk did not also find**, while **123 sectors are found only by the walk**. More samples of the same measure do not close the gap — and this claim can fail, which the saturated-bin comparison cannot.

5.3 Sectors with no known construction

Of the walk’s 283 sectors, 24 are of the form $\pm n^2$ and thus have the closed-form representative of §2.3; **259 do not**. The largest $|\mathcal{C}|$ realised is 192, which is *below* the best explicit construction (256) — the walk does not extend the known range of $|\mathcal{C}|$. What it produces is 259 distinct Chern numbers for which no parameter point was previously known.

6 Is this discovery?

The walk is label-free; the count is post-hoc. Nothing in Eq. (6) reads $\mathcal{C}$. Chern numbers are evaluated afterwards, by us, for both methods alike — so the comparison is fair, but the system is not *aware*, in real time, that it has entered a new sector.

What it produces is nevertheless new. By §2.3 there is no list to check off: for 92% of the sectors it realises, no construction is known. In that operational sense the result *is* discovery, in the way a

sector volume	sectors in bin	explorer	random	ratio
1–10%	33	33.0	33.0	1.0×
0.1–1%	26	26.0	26.0	1.0×
0.01–0.1%	29	29.0	29.0	1.0×
10^{-5} – 10^{-4}	30	30.0	29.4	1.0×
$< 10^{-5}$	37	33.4	17.2	1.9×
never seen in 10^6 samples	—	56.2	3.2	17.6×

Table 4: Mean over five seeds at 3×10^5 .

source	queries/samples	sectors	$\mathcal{C}$ range
all random sampling (10^6 Gaussian + $5 \times 3 \times 10^5$ baselines)	2.5×10^6	160	$[-86, 108]$
explorer, five runs	1.5×10^6	283	$[-192, 188]$
union		283	$[-192, 188]$

survey instrument discovers an object its pipeline identifies after the exposure. What is missing is not novelty but *online awareness*.

That gap is closable here, though not as cheaply as an earlier draft claimed. The Chern number is computable from the state alone — but not from the state the autoencoder sees: FHS on the 32×32 encoder input reproduces the converged label on only 9% of points. Closing the loop requires the separate 768×768 field, which the walk already computes and pays for, at 98ms per step. An explorer that closed this loop could *use* its findings, repelling from the set of sectors already visited rather than from its own past positions. We note that this would make the method *label-aware* and would require re-establishing the comparison against an active-learning baseline rather than against random sampling.

7 Limits

- **Validated at 50 parameters.** At 20 and 32 parameters the explorer loses $(-4, -1)$. At 98 the per-seed margin runs from **-49 to $+160$** across 5–8 seeds depending on budget and **no claim is supportable**; at 200 the walk stalls every 40–50 steps (390–487 re-initialisations against one at 50 parameters), traced to a stall criterion measured in the ℓ_∞ norm and therefore tightening as $\sqrt{d}$. At $d = 200$ the un-budget-checked bootstrap can also overshoot the nominal budget by $\approx 1.4 \times 10^4$ queries; at $d = 50$ the measured overshoot is zero.
- **Labels carry $\approx 2\%$ residual error.** $M = 768$ agrees with $M = 1536$ on 98.2% of points. The error is symmetric between methods (§2.4) but the absolute counts inherit it.
- **“Never seen in 10^6 ” is a membership criterion, not a volume bound.** It means volume $\lesssim 3 \times 10^{-6}$. Random sampling itself lands in this bin 3.2 times per run, so it is a resolution boundary, not a wall.
- **Rarity is defined under the Gaussian measure.** Another measure assigns different volumes. The comparison is self-consistent because Gaussian sampling *is* the baseline, but the statement is relative to it.
- **A query is an evaluation, not a hard solve**, and the walk’s gradient is free here (§3.3). The budget is a proxy for solver calls in an interacting problem, not a measured wall-clock cost, and solver cost is out of scope.

- **The winning configuration lacks a mechanism.** Step size 1.24 ($25\times$ the hand-chosen default), repulsion weight $w = 0.73$ and length $s = 0.083$, trail horizon 114 steps, one solve per 5 parameter steps, bootstrap radius $r = 0.087$, block length 319, gradient clip 1.01. Why large steps with *short*-range repulsion and a *compact* bootstrap cloud should work is unexplained; one-at-a-time scans had settled on $s = 0.5$ and one solve per 30 steps, which is worse.
- **Bracket outputs were not retained.** The Hyperband logs are not archived, so the selection history cannot be independently re-derived from stored artefacts.

References

- [1] J. Carrasquilla and R. G. Melko, *Machine learning phases of matter*, Nat. Phys. **13**, 431 (2017).
- [2] E. P. L. van Nieuwenburg, Y.-H. Liu, S. D. Huber, *Learning phase transitions by confusion*, Nat. Phys. **13**, 435 (2017).
- [3] Y. Zhang and E.-A. Kim, *Quantum loop topography for machine learning*, Phys. Rev. Lett. **118**, 216401 (2017).
- [4] J. F. Rodriguez-Nieva and M. S. Scheurer, *Identifying topological order through unsupervised machine learning*, Nat. Phys. **15**, 790 (2019).
- [5] D. J. Thouless, M. Kohmoto, M. P. Nightingale, M. den Nijs, Phys. Rev. Lett. **49**, 405 (1982).
- [6] F. D. M. Haldane, Phys. Rev. Lett. **61**, 2015 (1988).
- [7] X.-L. Qi, Y.-S. Wu, S.-C. Zhang, Phys. Rev. B **74**, 085308 (2006).
- [8] T. Fukui, Y. Hatsugai, H. Suzuki, *Chern numbers in discretized Brillouin zone*, J. Phys. Soc. Jpn. **74**, 1674 (2005).
- [9] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, A. Talwalkar, *Hyperband*, J. Mach. Learn. Res. **18**, 1 (2018).
- [10] Y. Zhang, *Autonomous discovery of symmetry-protected topological phases: benchmark results*, companion report (2026).